

Name: M. Abdallah Siddiqui
Roll No.: 23I-0617
Section: F

(Q:1)

$$L(A) = A^T$$

$$L(A) = \lambda A^T$$

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\lambda A^T = \lambda \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} \lambda a & \lambda c \\ \lambda b & \lambda d \end{bmatrix}$$

$$L(A) = \begin{bmatrix} \lambda a & \lambda c \\ \lambda b & \lambda d \end{bmatrix} \Rightarrow \begin{aligned} \lambda a &= a \\ \lambda b &= c \\ \lambda c &= b \\ \lambda d &= d \end{aligned}$$

Since $A = A^T$ and $A = -A^T$ for skew & sym matrices possible eigenvalues can be $\lambda = \pm 1$

$$\lambda = 1$$

$$\begin{bmatrix} a & b \\ b & d \end{bmatrix} \text{ or } \begin{bmatrix} a & c \\ c & d \end{bmatrix}$$

$$\lambda = -1$$

$$\begin{bmatrix} -a & -c \\ -c & -d \end{bmatrix} \text{ or } \begin{bmatrix} -a & -b \\ -b & -d \end{bmatrix}$$

Since the number of eigenvectors are 2, they are diagonalizable.

$$\text{EigenBasis} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}$$

(Q:2)

$$D(f) = f'$$

Derivative of a function

$$Ax = \lambda x$$

$$f = e^x$$

$$f = e^{\lambda x}$$

apply derivative

$$D(f) = \lambda e^{\lambda x} \Rightarrow D(f) = \lambda f$$

Therefore λ can be any real number from the domain $-\infty$ to ∞ when transformation is applied.

(Q:3)

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$$

$$x = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} x_2$$

$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix} x_2$$

eigenvectors

$\lambda = 5$ or $\lambda = -1 \Rightarrow \text{Eigenvalues}$

$$P = \begin{bmatrix} 1 & -2 \\ 2 & 2 \end{bmatrix}$$

$$D = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A = P D P^{-1}$$

$$= \begin{bmatrix} 1 & -2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 1/6 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 2 \\ 10 & -2 \end{bmatrix} \begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 1/6 \end{bmatrix}$$

$$= \begin{bmatrix} 5/3 - 2/3 & 5/3 + 2/6 \\ 10/3 + 2/3 & 10/3 - 2/6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} = A$$

It is diagonalizable since P has n independent columns.

(Q:4)

$$\lambda = 0$$
$$x = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} x_1$$

$$\lambda = 1$$
$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_1$$

$$P = \begin{bmatrix} -1 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$\Rightarrow$ Not diagonalizable since it has no n number of independent columns.

(Q:5)

$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 4 & 3 \\ -4 & -6-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} & 2-\lambda(-6-\lambda(1-\lambda)+9) - 4(-4(1-\lambda)+9) + 3(-12 - (3(-6-\lambda))) = \\ & 2-\lambda(-6+6\lambda-\lambda+\lambda^2+9) - 4(-4+4\lambda+9) + 3(-12+18+3\lambda) = \\ & 2-\lambda(\lambda^2+5\lambda+3) - 4(5+4\lambda) + 3(6+3\lambda) = 0 \\ & 2\lambda^2+10\lambda+6-\lambda^3-5\lambda^2-3\lambda-20-16\lambda+18+9\lambda=0 \\ & -\lambda^3-3\lambda^2+4=0 \end{aligned}$$

$$\lambda^3 + 3\lambda^2 - 4 = 0$$

By Synthetic division

$$\begin{array}{r|rrrr} \lambda & 1 & 3 & 0 & -4 \\ & & 1 & 4 & 4 \\ \hline & 1 & 4 & 4 & 0 \end{array}$$

$$\lambda^2 + 4\lambda + 4 = 0$$

$$\lambda = 1 \text{ or } \lambda = -2$$

for $\lambda = 1$

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ -4 & -7 & -3 & 0 \\ 3 & 3 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 9 & 9 & 0 \\ 0 & -9 & -9 & 0 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x = \begin{bmatrix} x_3 \\ -x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} x_3$$

for $\lambda = -2$

$$\left[\begin{array}{ccc|c} 4 & 4 & 3 & 0 \\ -4 & -4 & -3 & 0 \\ 3 & 3 & 3 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 4 & 4 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 3 & 3 & 3 & 0 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 4 & 4 & 3 & 0 \\ 3 & 3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 3/4 & 0 \\ 3 & 3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 3/4 & 0 \\ 0 & 0 & 3/4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x = \begin{bmatrix} -x_2 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} x_2$$

There are only 2 eigen vectors and $n=3$
so not diagonalizable.